

QFN PACKAGE

Quad Flat No-Lead — Complete Assembly Guide

Structure • Types • Thermal Pad • Stencil Design • Soldering
Reflow • X-Ray Inspection • Rework • QFN vs QFP

PCBA Manufacturing & Advanced Packaging Standards

IPC-7093 | IPC/J-STD-001 | IPC-A-610 | JEDEC

1. QFN OVERVIEW

QFN (Quad Flat No-Lead) is a surface-mount IC package where the leads are replaced by metal pads on the bottom perimeter of the package. Unlike QFP (Quad Flat Package) which has gull-wing leads extending from the sides, QFN has no visible leads — only flat contact pads on the bottom edges and a large exposed thermal pad in the center.

Why QFN is Popular:

- **Compact size:** 30–50% smaller footprint than equivalent QFP
- **Excellent thermal performance:** Exposed thermal pad dissipates heat directly to PCB
- **Superior electrical performance:** Short lead lengths reduce inductance and resistance
- **Low profile:** 0.9–1.0 mm height typical (vs. 1.4 mm for TQFP)
- **Good solder joint reliability:** Large contact area and self-alignment during reflow
- **Cost-effective:** Simple leadframe construction; no gull-wing forming

QFN Challenges:

- **Hidden solder joints:** Peripheral pads are under the package — not visible for optical inspection
- **Thermal pad soldering:** Large center pad requires special stencil design to prevent floating
- **Rework difficulty:** No leads to grip; requires hot-air and skilled technique
- **Voiding:** Center thermal pad prone to solder voiding
- **Pad pitch:** Fine pitch (0.4–0.65 mm) requires precise stencil and placement

■ **In-House Note:** QFN is now the dominant package for power management ICs, motor drivers, RF transceivers, and small microcontrollers. In-house assembly handles QFN from 2×2 mm (single DFN) to 12×12 mm (64-pin QFN) on standard SMT lines.

2. QFN PACKAGE STRUCTURE

2.1 Cross-Section Anatomy

The QFN package consists of multiple layers that provide electrical connection, thermal dissipation, and mechanical protection.

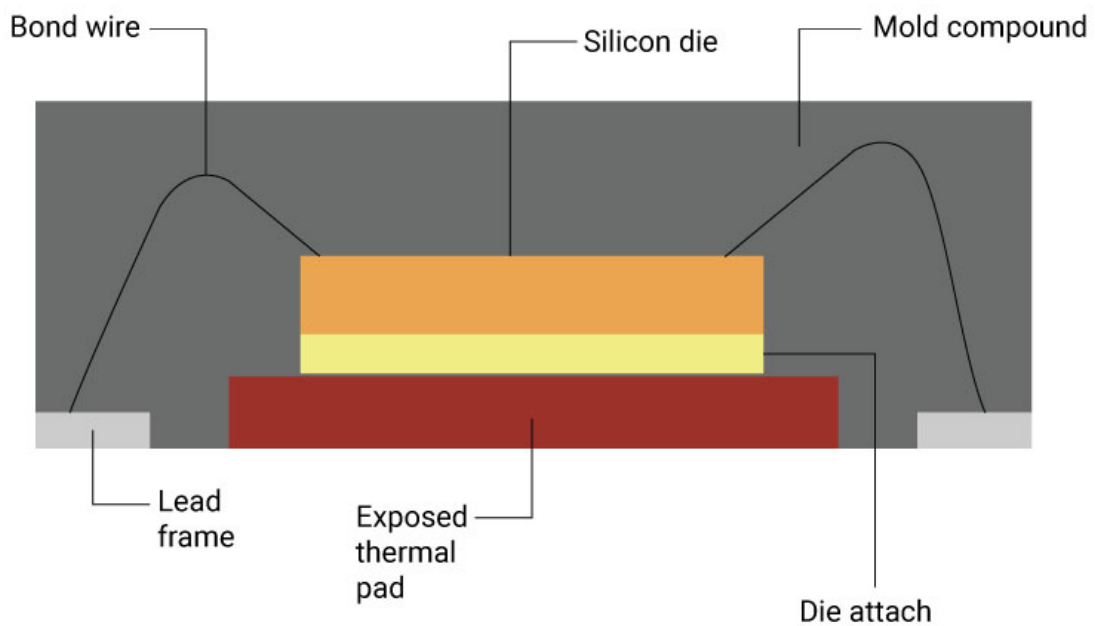


Figure 1: QFN Cross-Section — Bond Wire, Silicon Die, Mold Compound, Lead Frame, Exposed Thermal Pad, Die Attach

Key Components:

- **Silicon Die:** The active IC chip, typically 0.2–0.4 mm thick
- **Die Attach:** Conductive epoxy or solder attaching die to thermal pad
- **Bond Wires:** Gold or copper wires (25–35 μm diameter) connecting die pads to leadframe
- **Leadframe:** Copper alloy frame forming peripheral pads and thermal pad
- **Exposed Thermal Pad (E-pad):** Large central copper pad for heat dissipation and electrical grounding
- **Mold Compound:** Epoxy encapsulant protecting die and wire bonds
- **Plating:** Matte tin or NiPdAu on pads for solderability

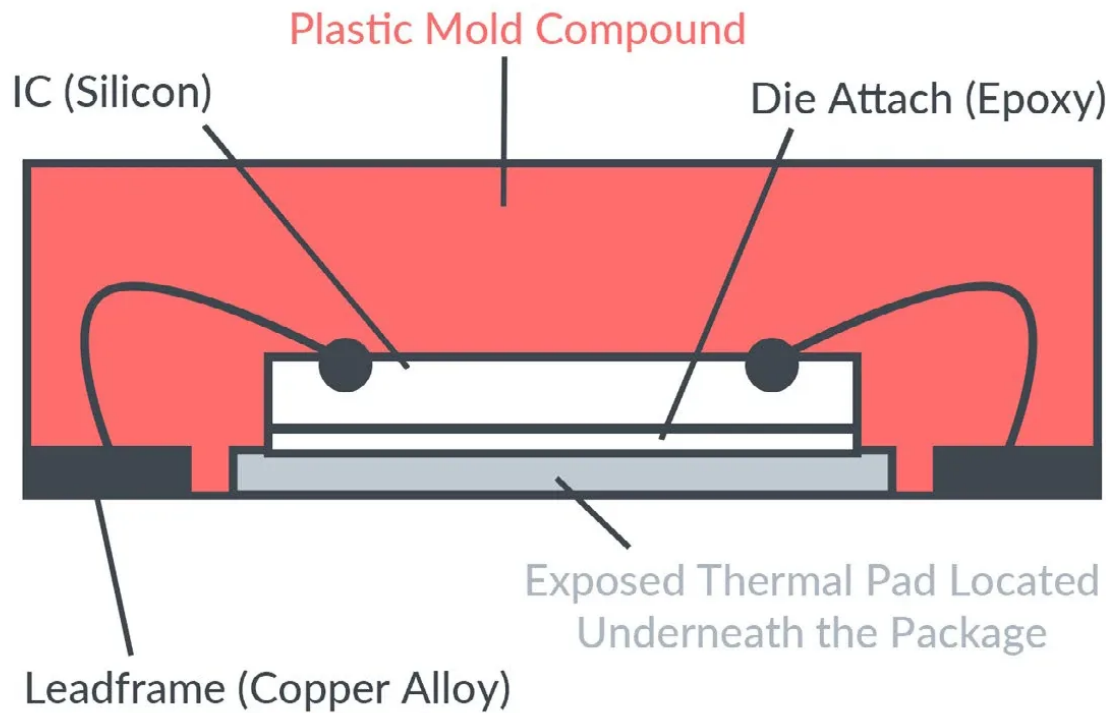


Figure 2: QFN Cross-Section — IC (Silicon), Plastic Mold Compound, Die Attach (Epoxy), Exposed Thermal Pad, Leadframe (Copper Alloy)

2.2 QFN vs QFP Comparison

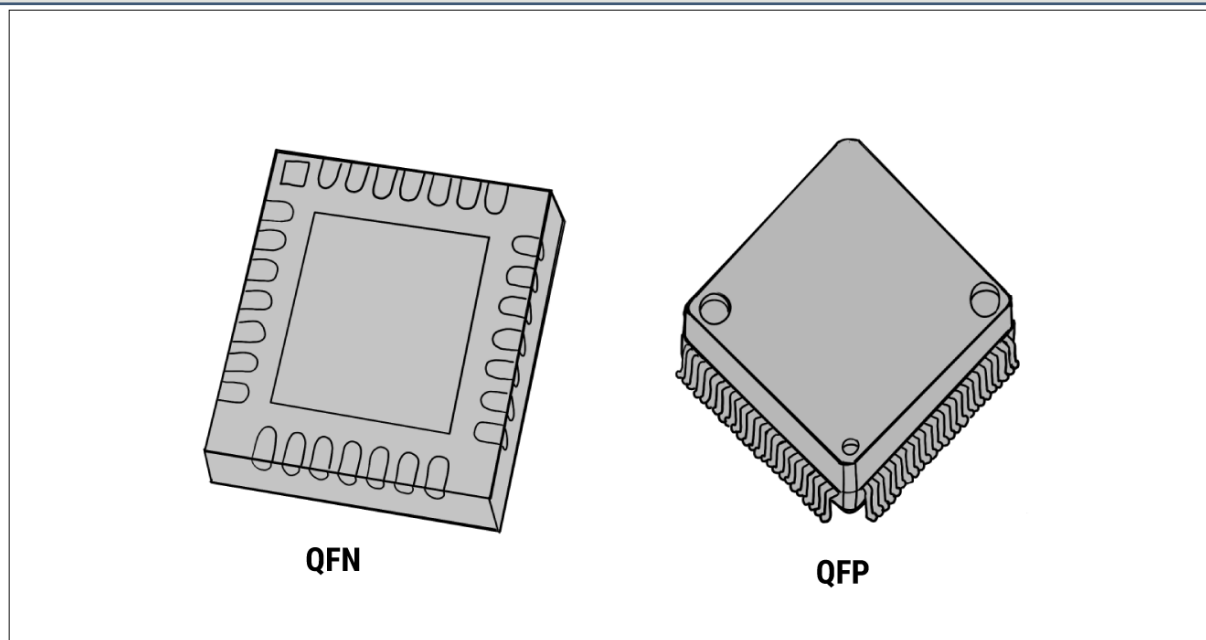


Figure 3: QFN vs QFP — QFN Has No Leads, QFP Has Gull-Wing Leads

The fundamental difference between QFN and QFP is the lead configuration. QFN has flat pads on the bottom perimeter with no leads extending beyond the package body. QFP has gull-wing leads that extend from all four sides, making them visible and inspectable.

Feature	QFN	QFP	Advantage
Footprint	Smaller (no leads)	Larger (leads extend)	QFN — space savings
Height	0.9–1.0 mm	1.4–1.6 mm	QFN — low profile

Thermal Performance	Excellent (E-pad)	Poor (no thermal pad)	QFN — direct heat path
Electrical Performance	Excellent (short paths)	Good (longer leads)	QFN — lower inductance
Visual Inspection	Difficult (hidden joints)	Easy (visible leads)	QFP — AOI capable
Rework	Difficult (no leads)	Easy (grip leads)	QFP — simpler rework
Self-Alignment	Moderate	Good (lead surface tension)	QFP — better alignment
Cost	Lower	Higher	QFN — cheaper
I/O Density	High	Moderate	QFN — more I/O per area

Table 1: QFN vs QFP Comparison

3. QFN PACKAGE TYPES

3.1 Standard QFN

Standard QFN is the most common variant with peripheral pads on all four sides and a central exposed thermal pad. Body sizes range from 3×3 mm to 12×12 mm.

- **Body size:** 3×3 mm to 12×12 mm
- **Lead count:** 8 to 88+ pads
- **Pad pitch:** 0.4 mm, 0.5 mm, 0.65 mm, 0.8 mm, 1.0 mm
- **Thermal pad:** Present in center
- **Height:** 0.9–1.0 mm

3.2 DFN (Dual Flat No-Lead)

DFN is a two-sided variant of QFN with pads only on two opposite sides (like SOIC without leads). Used for simple devices with few I/O pins.

- **Body size:** 2×2 mm to 6×6 mm
- **Lead count:** 2 to 16 pads
- **Pad pitch:** 0.5 mm, 0.65 mm, 0.95 mm
- **Thermal pad:** Optional (small DFN may omit)
- **Use:** MOSFETs, LDO regulators, small logic devices

3.3 Power QFN (PQFN)

PQFN is optimized for power devices with enhanced thermal performance. The thermal pad is significantly larger relative to the package body.

- **Body size:** 3×3 mm to 8×8 mm
- **Thermal pad:** Up to 80% of package area
- **Lead count:** 8 to 24 pads
- **Current rating:** Up to 100A for power MOSFETs
- **Use:** Power MOSFETs, motor drivers, DC-DC converters

3.4 Wettable Flank QFN

Wettable flank QFN has specially treated side surfaces that allow solder to wet the package sidewall. This creates a visible solder fillet for optical inspection — solving the hidden joint problem of standard QFN.

- **Side treatment:** Chemically etched or plated to promote solder wetting
- **Inspection:** AOI capable — visible fillet on side
- **Reliability:** Improved solder joint strength
- **Cost:** Slightly higher than standard QFN
- **Use:** Automotive (Tesla, Zoox) where AOI is mandatory

3.5 Air-Cavity QFN

Air-cavity QFN has an open cavity on top (no mold compound over the die) for RF and optical applications where the die must be exposed.

- **Construction:** Die attached to thermal pad; wire bonds exposed
- **Protection:** Requires lid or glob-top encapsulation after assembly
- **Use:** RF power amplifiers, LED drivers, optical sensors
- **Handling:** Extreme ESD sensitivity — die is exposed

3.6 Multi-Row QFN

Multi-row QFN has two or more rows of pads per side, dramatically increasing I/O count for complex devices.

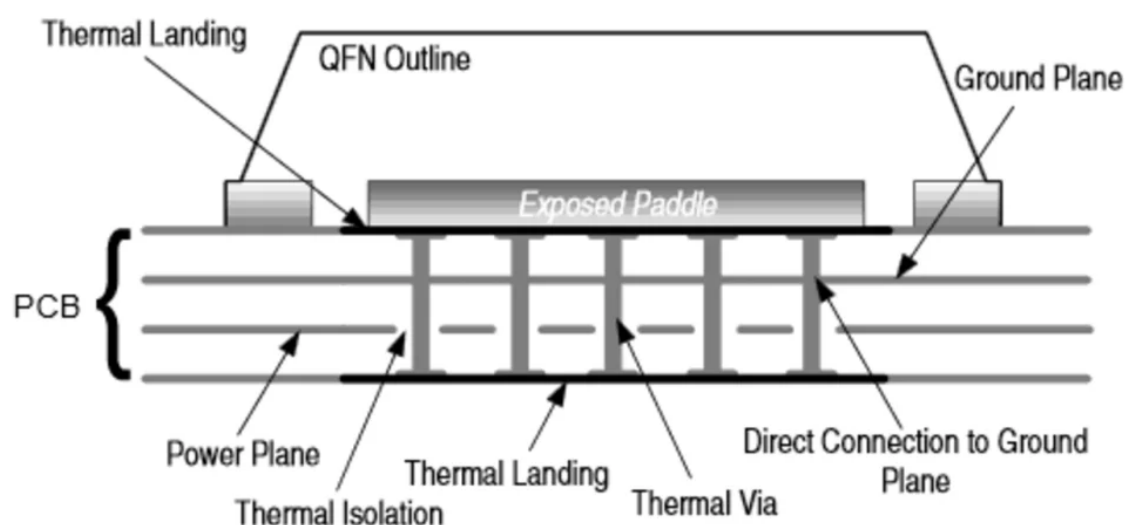
- **Rows:** 2 or 3 rows of pads per side
- **I/O count:** Up to 200+ pads in compact body
- **Pad pitch:** 0.4 mm typical
- **Routing:** Requires microvias and advanced PCB technology
- **Use:** Complex SoC, FPGA, high-speed processors

4. EXPOSED THERMAL PAD (E-PAD)

4.1 Purpose of the Thermal Pad

The **exposed thermal pad** (also called E-pad, exposed paddle, or ground pad) is the defining feature of QFN packages. It serves dual functions:

- **Thermal dissipation:** Provides low thermal resistance path from die to PCB
- **Electrical grounding:** Often connected to IC ground for EMI reduction and stable reference
- **Mechanical stability:** Large solder area prevents package tilt or floating during reflow



Side View

Figure 4: QFN Thermal Pad Cross-Section — Thermal Landing, Thermal Vias, Ground Plane, Power Plane

4.2 PCB Thermal Via Design

Thermal vias connect the top-layer thermal pad to internal ground planes, spreading heat throughout the PCB.

- **Via diameter:** 0.2–0.3 mm (8–12 mil)
- **Via pitch:** 0.8–1.2 mm (center-to-center)
- **Via count:** 4–25 vias depending on thermal pad size
- **Via fill:** Solder-filled or epoxy-filled (prevent solder wicking)
- **Via tenting:** Solder mask over vias on bottom side (prevent solder loss)
- **Ground connection:** Vias connect to solid ground plane on inner layers

Thermal Resistance Comparison:

- **QFN with thermal vias:** $\theta_{JA} = 15\text{--}30^{\circ}\text{C/W}$
- **QFN without thermal vias:** $\theta_{JA} = 40\text{--}60^{\circ}\text{C/W}$
- **QFP (no thermal pad):** $\theta_{JA} = 50\text{--}80^{\circ}\text{C/W}$
- **Improvement:** Thermal vias reduce junction temperature by 20–40°C

4.3 Thermal Pad Soldering Challenges

The large thermal pad presents unique soldering challenges:

1. Solder Voiding:

Air pockets trapped under the thermal pad reduce heat transfer. Target: < 25% void area.

→ **Solution:** Window pane stencil design; multiple small apertures instead of one large opening

2. Package Floating:

Excessive solder paste on thermal pad lifts the package, preventing peripheral pad contact.

→ **Solution:** Reduce thermal pad paste coverage to 50–70% of pad area

→ **Solution:** Use 0.125 mm (5 mil) thick stencil for thermal pad

3. Solder Wicking into Vias:

Molten solder flows into unfilled thermal vias, starving the thermal pad joint.

→ **Solution:** Use solder mask-defined (SMD) vias or epoxy-filled vias

■ ■ **Critical:** Never leave the thermal pad unconnected (floating). A floating thermal pad acts as an antenna, causing EMI issues and unpredictable device behavior. Always connect to ground (or power, per datasheet) with thermal vias.

5. QFN STENCIL DESIGN

5.1 Peripheral Pad Stencil

The peripheral pads require standard stencil design with attention to pitch and aperture geometry.

- **Stencil thickness:** 100–150 μm (standard); 80–100 μm for 0.4 mm pitch
- **Aperture shape:** Rectangular with rounded corners
- **Aperture size:** 1:1 to pad size (no oversize for QFN)
- **Area ratio:** ≥ 0.66 (aperture area / wall area)
- **Electroformed stencil:** Recommended for 0.4–0.5 mm pitch

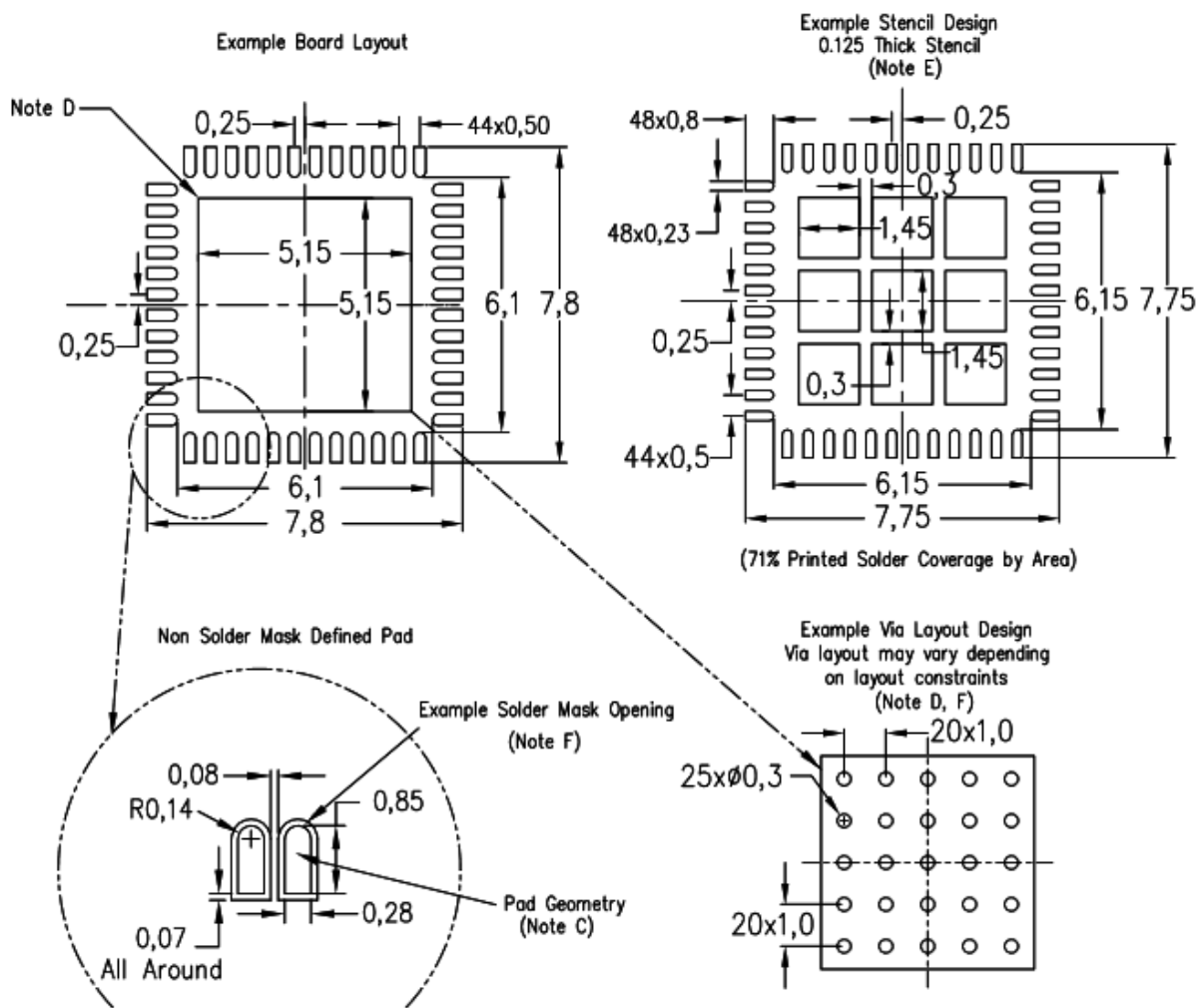


Figure 5: QFN Footprint, Stencil Design, and Thermal Via Layout — 71% Printed Solder Coverage

5.2 Thermal Pad Stencil Design

The thermal pad stencil is the most critical aspect of QFN assembly. Incorrect design causes voiding, floating, or insufficient solder.

Window Pane Design (Recommended):

- Divide thermal pad aperture into multiple small rectangles or squares
- **Individual aperture size:** 1.0–2.0 mm per segment
- **Web thickness between apertures:** 0.2–0.3 mm

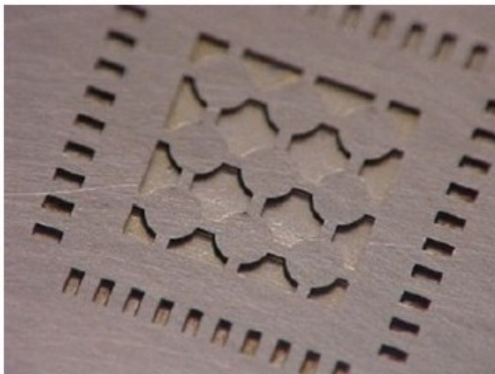
- **Total coverage:** 50–70% of thermal pad area
- **Benefit:** Reduces voiding; prevents floating; improves outgassing

Round Dot Design (Alternative):

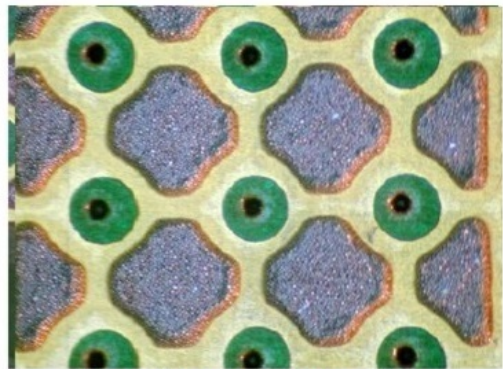
- Array of small round apertures (0.5–1.0 mm diameter)
- **Coverage:** 50–60% of thermal pad area
- **Benefit:** Simple laser cutting; good for small thermal pads

Cross-Hatch Design:

- Grid pattern with 1.0–1.5 mm squares
- **Coverage:** 60–70% of thermal pad area
- **Benefit:** Balanced solder distribution; easy inspection



Stencil with specially designed Apertures to avoid vias



Resulting print of solder paste onto pads

Figure 6: QFN Stencil Apertures — Specially Designed to Avoid Vias (Left) and Resulting Solder Print (Right)

5.3 Stencil Design Rules Summary

Parameter	Peripheral Pads	Thermal Pad
Stencil Thickness	100–150 μm	100–150 μm (or 0.125 mm step-down)
Aperture Shape	Rounded rectangle	Window pane / Grid / Round dots
Aperture Size	1:1 to pad	1.0–2.0 mm segments
Coverage	100%	50–70% of pad area
Area Ratio	≥ 0.66	≥ 0.66 per segment
Via Avoidance	N/A	Aperture offset from vias

Table 2: QFN Stencil Design Rules

6. QFN SOLDERING & REFLOW

6.1 Solder Paste

QFN soldering requires paste matched to the package plating and application requirements.

- **Alloy:** SAC305 (standard lead-free) or SAC387 (higher reliability)
- **Powder type:** Type III (25–45 µm) for 0.5+ mm pitch; Type IV (20–38 µm) for 0.4 mm pitch
- **Flux:** No-clean ORH1 (standard) or ORH0 (low residue)
- **Activity:** ROL1 or ORH1 for OSP and NiPdAu surfaces

6.2 Reflow Profile

QFN reflow follows standard SAC305 profiles with emphasis on even heating to prevent package tilt and ensure thermal pad wetting.

Standard SAC305 QFN Profile:

- **Preheat:** 25 → 150°C at 1.5–2.5°C/sec
- **Soak:** 150 → 180°C for 60–90 seconds
- **Ramp to peak:** 180 → 245°C at 1.5–2.0°C/sec
- **Peak:** 245–250°C
- **TAL:** 60–90 seconds above 217°C
- **Cooling:** 2–4°C/sec down to 100°C

QFN-Specific Reflow Considerations:

- **Even heating:** Ensure all sides reach liquidus simultaneously to prevent tilt
- **Thermal mass:** Large QFN (> 10×10 mm) requires longer TAL
- **Nitrogen:** Recommended for 0.4–0.5 mm pitch; improves wetting and reduces voids
- **Board support:** Use pins for large QFN to prevent warpage
- **Conveyor speed:** Reduce for high-density boards with multiple QFNs

6.3 Self-Alignment

QFN has **limited self-alignment** compared to QFP or BGA because the peripheral pads are flat and have less surface tension. However, the thermal pad provides some self-centering if paste volume is correct.

- **Self-alignment limit:** ~25% of pad pitch (less than QFP's 50%)
- **Example:** 0.5 mm pitch QFN tolerates ± 0.125 mm placement offset
- **Requirement:** Tighter placement accuracy than QFP — ± 25 µm at 3σ

6.4 Common Soldering Defects

1. Thermal Pad Voiding

Cause: Insufficient flux outgassing, large single aperture, no vias

→ **Fix:** Window pane stencil; solder-filled vias; nitrogen atmosphere

2. Package Floating / Tilt

Cause: Excessive paste on thermal pad; uneven heating

→ **Fix:** Reduce thermal pad coverage to 50–70%; verify even oven heating

3. Insufficient Peripheral Solder

Cause: Thermal pad solder wicking starves peripheral joints

→ **Fix:** Reduce thermal pad paste; verify stencil alignment

4. Bridging (Fine Pitch)

Cause: Excessive paste volume; misaligned print

→ **Fix:** Reduce aperture by 10%; verify 1:1 pad match

5. Head-in-Pillow (Rare in QFN)

Cause: Oxidation on pad or package; insufficient flux

→ **Fix:** Fresh paste; nitrogen; higher activity flux

7. QFN INSPECTION METHODS

7.1 Visual Inspection Limitations

Standard optical inspection has limited capability for QFN because the solder joints are hidden under the package body.

- **Peripheral pads:** Only the outer edge is visible; solder wetting under package is hidden
- **Thermal pad:** Completely hidden — no visual access
- **What AOI can detect:** Presence, orientation, gross displacement, solder paste after print
- **What AOI cannot detect:** Voiding, head-in-pillow, insufficient wetting under package

7.2 X-Ray Inspection

X-ray is the primary method for verifying QFN solder joint quality, especially the thermal pad.

2D X-Ray:

- Detects: Thermal pad voiding, peripheral pad presence, gross bridging
- Limitation: Superimposed images; cannot distinguish layers
- Cost: Lower than 3D X-ray

3D X-Ray (AXI):

- Detects: Void volume, individual pad analysis, crack propagation
- Advantage: Volumetric reconstruction of each joint
- Cost: High — premium equipment

7.3 Voiding Acceptance Criteria

Thermal pad voiding is the most common QFN defect detected by X-ray.

- **IPC-A-610 Class 2:** $\leq 25\%$ void area per thermal pad
- **IPC-A-610 Class 3:** $\leq 15\%$ void area per thermal pad
- **Automotive (Tesla, Zook):** $\leq 10\%$ void area (customer-specific)
- **Power QFN:** $\leq 20\%$ void area (thermal performance critical)

7.4 Electrical Testing

- **Continuity:** Verify all ground pins connected to thermal pad
- **Isolation:** Verify no shorts between adjacent pins
- **Functional:** Power-on test; verify IC operates within specification
- **Thermal:** IR camera measurement of case temperature under load

■ **In-House Best Practice:** For automotive QFN assemblies, implement 100% X-ray inspection on all QFN devices. Record void percentage by lot number and device position. Establish SPC control limits for voiding trends.

8. QFN REWORK PROCESS

8.1 Rework Challenges

QFN rework is more difficult than QFP because there are no leads to grip or heat. The package sits flat on the PCB, and the thermal pad creates a large heat sink.

- **No leads:** Cannot use tweezers on gull-wing leads
- **Large thermal pad:** Requires significant heat to melt solder
- **Hidden joints:** Cannot see when solder is molten
- **Pad damage risk:** Excessive heat or force lifts PCB pads

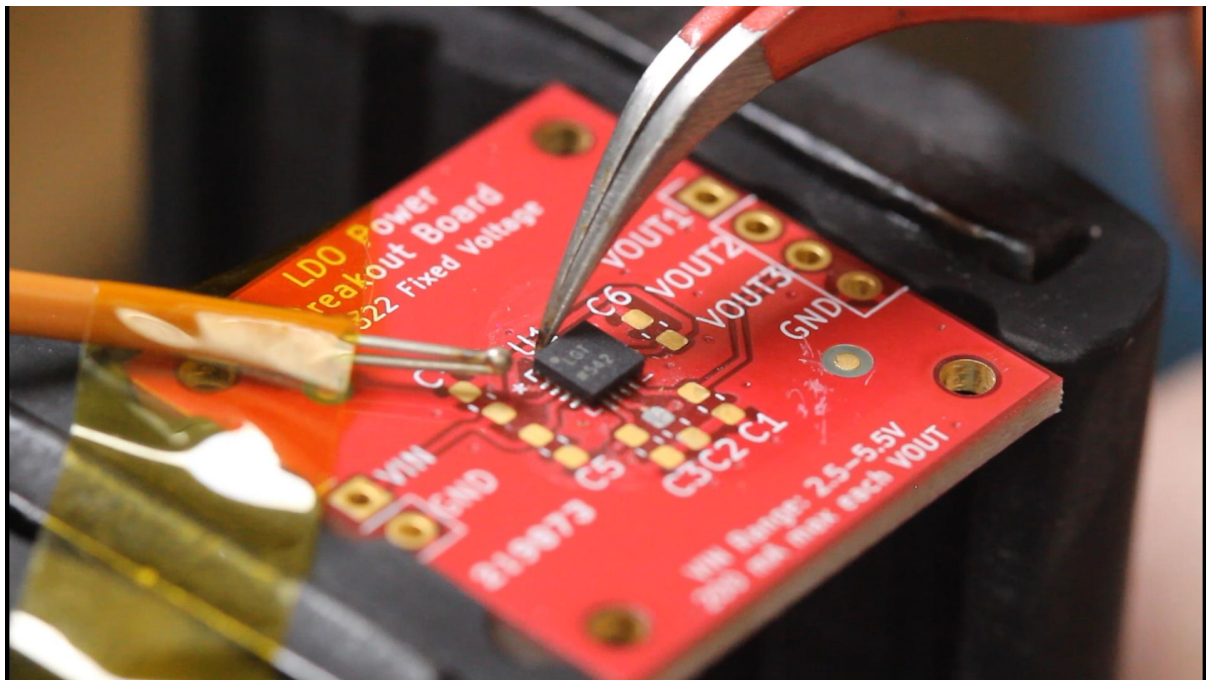


Figure 7: QFN Rework — Hot Air Station with Tweezers Placement on PCB

8.2 Rework Equipment

- **Hot-air rework station:** 800–1500W with programmable profile
- **Nozzle:** Custom size matching QFN body (2×2 mm to 12×12 mm)
- **Preheater:** Bottom-side board heating (100–150°C)
- **Vacuum pickup:** For component removal and replacement
- **Soldering iron:** For pad cleaning and touch-up
- **Solder paste:** Dispense syringe for applying fresh paste

8.3 Rework Procedure

1. **Preparation:** Secure board in fixture; apply Kapton tape to protect nearby components
2. **Preheat:** Heat board from bottom to 100–150°C for 2–3 minutes
3. **Apply flux:** Liquid flux around QFN perimeter and thermal pad area
4. **Heat with hot air:** 300–350°C air, 3–5 mm distance, circular motion over package
5. **Remove:** When solder melts (package shifts slightly), lift with vacuum tool

6. **Clean pads:** Use solder wick and flux to remove old solder; verify pad flatness
7. **Apply paste:** Dispense fresh SAC305 paste on all pads (peripheral + thermal)
8. **Place new QFN:** Align using vision system or manual under microscope
9. **Reflow:** Apply hot air following standard profile; allow self-alignment
10. **Cool:** Natural cooling; do not force cool
11. **Inspect:** X-ray verification; electrical test

8.4 Pad Repair

If pads are lifted or damaged during rework:

- **Minor damage:** Clean pad; apply conductive epoxy; re-solder
- **Major damage:** Run jumper wire to nearest via or trace
- **Thermal pad damage:** May require board scrap if ground plane connection lost
- **Prevention:** Proper preheat reduces pad lifting risk

■■ **Warning:** QFN rework has high risk of pad lifting, especially on thin PCBs (< 1.0 mm) or boards with OSP finish. Always use bottom preheater and minimize dwell time. If pad lifting occurs on > 2 pads, scrap the board.

9. QFN PACKAGE COMPLETE MATRIX

QFN Type	Body Size	Pins	Pitch	Thermal Pad	Typical Use
Standard QFN	3×3 to 12×12 mm	8–88+	0.4–1.0 mm	Yes	MCU, PMIC, RF
DFN	2×2 to 6×6 mm	2–16	0.5–0.95 mm	Optional	MOSFET, LDO, logic
PQFN	3×3 to 8×8 mm	8–24	0.5–1.27 mm	Large (80% area)	Power MOSFET, motor driver
Wettable Flank	3×3 to 10×10 mm	16–64	0.4–0.65 mm	Yes	Automotive AOI
Air-Cavity	4×4 to 8×8 mm	16–48	0.5–0.8 mm	Yes (exposed die)	RF PA, LED driver
Multi-Row	5×5 to 12×12 mm	48–200+	0.4 mm	Yes	SoC, FPGA, processor

Table 3: QFN Package Types Complete Matrix

10. QUICK REFERENCE SUMMARY

QFN Structure:

- **Silicon die** on **exposed thermal pad** with **bond wires** to **peripheral pads**
- **No leads** — flat pads on bottom perimeter
- **Mold compound** encapsulates die and wire bonds
- **Plating:** Matte tin or NiPdAu on pads

QFN vs QFP:

- **QFN:** Smaller, better thermal, better electrical, lower profile, lower cost
- **QFP:** Easier inspection, easier rework, better self-alignment
- **Winner:** QFN for new designs; QFP for legacy/rework-heavy

QFN Types:

- **Standard QFN:** 3×3 to 12×12 mm, 8–88 pins, 0.4–1.0 mm pitch
- **DFN:** 2-sided, 2–16 pins, MOSFET/LDO
- **PQFN:** Large thermal pad (80% area), power devices
- **Wettable Flank:** Sides treated for AOI visibility, automotive
- **Air-Cavity:** Exposed die, RF/optical, ESD sensitive
- **Multi-Row:** 2–3 rows, 200+ pins, complex SoC

Thermal Pad (E-Pad):

- **Purpose:** Heat dissipation + electrical grounding + mechanical stability
- **Via design:** 0.2–0.3 mm diameter, 0.8–1.2 mm pitch, 4–25 vias
- **Via fill:** Solder-filled or epoxy-filled (prevent wicking)
- **Never leave floating** — connect to ground (or per datasheet)
- **Thermal resistance:** 15–30°C/W with vias; 40–60°C/W without

Stencil Design:

- **Peripheral pads:** 1:1 aperture, rounded rectangle, ≥ 0.66 area ratio
- **Thermal pad:** Window pane / grid / round dots, 50–70% coverage
- **Thickness:** 100–150 μm standard; 80–100 μm for 0.4 mm pitch
- **Electroformed:** Recommended for fine pitch

Reflow:

- **Profile:** Standard SAC305; peak 245–250°C
- **TAL:** 60–90 seconds above 217°C
- **Nitrogen:** Recommended for 0.4–0.5 mm pitch
- **Self-alignment:** ~25% of pad pitch (tighter than QFP)

→ **Placement accuracy:** $\pm 25\text{ }\mu\text{m}$ at 3σ

Inspection:

- **AOI:** Limited — only presence, orientation, paste print
- **X-ray 2D:** Voiding, bridging, pad presence
- **X-ray 3D (AXI):** Void volume, individual pad analysis
- **Void limits:** $\leq 25\%$ (Class 2); $\leq 15\%$ (Class 3); $\leq 10\%$ (Automotive)

Rework:

- **Equipment:** Hot-air station + bottom preheater + custom nozzle
- **Preheat:** 100–150°C bottom, 2–3 minutes
- **Hot air:** 300–350°C, 3–5 mm distance, circular motion
- **Pad cleaning:** Solder wick + flux; verify flatness
- **Risk:** Pad lifting on thin PCBs — use preheat to minimize

Common Defects:

- **Thermal pad voiding:** Window pane stencil, nitrogen, filled vias
- **Package floating:** Reduce thermal pad coverage to 50–70%
- **Insufficient peripheral solder:** Reduce thermal pad paste volume
- **Bridging:** Reduce aperture 10%, verify alignment
- **Pad lifting (rework):** Preheat, minimize dwell, gentle handling

References & Standards:

- IPC-7093: Design and Assembly Process Implementation for Bottom Termination Components
- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- IPC-A-610: Acceptability of Electronic Assemblies
- JEDEC MO-220: Plastic Quad Flat No-Lead Package (QFN) Family
- JEDEC MO-229: Very Very Thin Profile Fine Pitch Quad Flat No-Lead (VFQFN)
- IPC-7527: Stencil Design Guidelines
- IPC-1752A: Materials Declaration Management
- Manufacturer Guidelines: Texas Instruments, Analog Devices, NXP, Microchip, STMicroelectronics